

HR Analytics – Employee Attrition & Performance Analysis

Project Documentation | Final Version

This documentation is based on the supplied cleaned dataset, SQL script, Python EDA notebook, Power BI dashboard export, and Tableau packaged workbook. The cleaned dataset contains 1,470 employee records.

1. Project Overview

HR Analytics – Employee Attrition & Performance Analysis analyzes employee workforce composition and attrition using SQL Server, Python, Power BI, and Tableau.

End-to-End Workflow:

Raw HR Data → SQL Server – Data Preparation & Analysis → Python EDA – Cleaning & Exploratory Analysis → Power BI – Interactive Business Intelligence → Tableau – Interactive HR Dashboards → Business-Focused Attrition Insights

2. Business Objectives

- Measure total workforce and employee attrition.
- Calculate overall attrition rate.
- Identify departments and job roles with comparatively higher attrition.
- Analyze attrition by age, gender, salary, overtime, satisfaction, work-life balance, and business travel.
- Analyze employee tenure and retention patterns.
- Evaluate relationships between compensation, performance, and employee turnover.
- Provide an interactive framework for HR decision-making.

3. Dataset Profile

Item	Verified Value
Rows	1,470
Columns	34
Age Range	18–60 years
Monthly Income Range	1,009–19,999
Attrition Encoding	0 = Active, 1 = Left

4. Core Workforce Metrics

Metric	Value
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Total Employees	1,470
Active Employees	1,233
Attrition Employees	237
Attrition Rate	16.12%
Average Age	36.92
Average Monthly Income	6,502.93
Average Years at Company	7.01
Average Job Satisfaction	2.73
Average Work-Life Balance	2.76
Average Performance Rating	3.15

5. SQL Server – Data Preparation & Analysis

SQL Server was used for data preparation, validation, KPI calculation, and advanced analytical queries.

- Data validation and record/duplicate checks.
- Missing-value checks.
- Removal of unnecessary fields: EmployeeCount, Over18, and StandardHours.
- Creation/use of binary Attrition flag.
- Core workforce and attrition KPI calculations.
- Department and job-role analysis.
- Demographic analysis.
- Overtime and satisfaction analysis.
- Business-travel analysis.
- Experience, promotion, performance, and salary-hike analysis.
- Risk and salary ranking analysis.

Advanced SQL Concepts:

- SELECT
- WHERE
- GROUP BY
- CASE
- Aggregate Functions
- Conditional Aggregation
- CTEs
- Window Functions
- Ranking Functions

6. Python – Exploratory Data Analysis

Python was used for data cleaning, validation, exploratory analysis, visualization, and cleaned dataset preparation.

Libraries Used:

- Pandas
- NumPy
- Matplotlib
- Seaborn
- Jupyter Notebook

EDA Activities:

- Loaded and inspected the dataset.
- Checked dataset structure.
- Performed data cleaning and validation.
- Analyzed overall employee attrition.
- Analyzed department-wise attrition rate.
- Examined age distribution by attrition.
- Analyzed monthly income distributions.
- Studied relationships between numerical variables.
- Created correlation analysis.

Visualizations Created:

- Overall Attrition Visualization
- Department-wise Attrition Rate
- Age Distribution by Attrition
- Monthly Income Box Plot
- Correlation Heatmap

Python Files: HR_EDA.ipynb; cleaned_data.csv

7. Power BI – Dashboard Documentation

The Power BI solution contains three analytical dashboard pages plus a Data Dictionary & Metric Documentation page.

Dashboard 1 – Executive HR Overview

Key Metrics:

- Total Employees: 1.47K
- Average Monthly Income: 6,502.93
- Average Age: 37
- Attrition Rate: 16.12%
- Attrition Count: 237

Analysis:

- Department attrition
- Gender distribution
- Job satisfaction by department
- Attrition by age group
- Monthly income vs attrition
- Overtime vs attrition

Filters:

- Department
- Gender
- Education Field
- Attrition

Dashboard 2 – Department Analysis

Key Metrics:

- Total Employees: 1,470
- Attrition Count: 237
- Active Employees: 1,233
- Average Years at Company: 7.01
- Average Job Satisfaction: 2.73

Analysis:

- Attrition by department
- Education field distribution
- Average salary by department
- Department summary
- Employees by department

Filters:

- Department
- Gender
- Attrition
- Education Field
- Job Role

Dashboard 3 – Employee Insights

Key Metrics:

- Total Employees: 1.47K
- Average Monthly Income: 6,502.93
- Average Age: 37

- Attrition Rate: 16.12%
- Attrition Count: 237

Analysis:

- Department and role attrition summary
- Age group distribution
- Attrition rate by job role
- Employee engagement score
- Attrition rate by monthly income

Filters:

- Department
- Gender
- Overtime

8. Power BI Verified Department Results

Department	Employees	Active	Attrition	Rate
Research & Development	961	828	133	13.84%
Sales	446	354	92	20.63%
Human Resources	63	51	12	19.05%
Total	1,470	1,233	237	16.12%

9. Tableau – Final Workbook

The packaged Tableau workbook contains two interactive dashboards and an embedded Hyper extract.

- Workforce Overview
- HR Attrition & Performance Analysis

10. Tableau Calculated Fields

- Total Employees – COUNTD(EmployeeNumber)
- Active Employees – distinct employees where Attrition = 0
- Attrition Employees – distinct employees where Attrition = 1
- Attrition Rate – Attrition Employees / Total Employees
- Average Age – AVG(Age)
- Average Monthly Income – AVG(MonthlyIncome)
- Average Years at Company – AVG(YearsAtCompany)

Age Group: 18–24, 25–34, 35–44, 45–54, 55+

Salary Band: Below 3K, 3K–6K, 6K–10K, 10K+

11. Tableau Dashboard 1 – Workforce Overview

Executive KPIs:

- Total Employees
- Active Employees
- Attrition Employees
- Attrition Rate
- Average Age
- Average Monthly Income
- Average Years at Company

7 Charts:

Analysis	Chart Type
Attrition by Department	Horizontal Bar
Attrition by Job Role	Horizontal Bar
Attrition by Age Group	Bar Chart
Attrition by Gender	Donut Chart
Attrition by Overtime	Bar Chart
Attrition by Job Satisfaction	Bar Chart
Attrition by Salary Band	Bar Chart

Filters: Department; Job Role; Gender; Overtime

12. Tableau Dashboard 2 – Attrition & Performance Analysis

KPIs:

- Total Employees
- Attrition Employees
- Attrition Rate
- Average Monthly Income
- Average Years at Company

6 Charts:

Analysis	Chart Type
Attrition Rate by Department	Horizontal Bar
Attrition Rate by Job Role	Horizontal Bar
Attrition by Salary Band	Column Bar
Attrition by Years at Company	Line Chart
Attrition by Work-Life Balance	Column Bar
Attrition by Business Travel	Bar Chart

Filters: Department; Job Role; Gender; Age Group; Overtime

13. Tableau Navigation & Formatting

- Dashboard 1 ↔ Dashboard 2 navigation bar.
- Highlighted background for the active dashboard.
- Consistent dashboard and chart titles.
- Formatted KPI cards.
- Organized chart rows and columns.
- Balanced spacing and alignment.
- Organized filters.
- Consistent visual hierarchy.
- Business-focused dashboard presentation.

14. Dataset-Derived Attrition Insights

- Human Resources: 19.05% attrition rate (12 of 63 employees).
- Research & Development: 13.84% attrition rate (133 of 961 employees).
- Sales: 20.63% attrition rate (92 of 446 employees).
- Sales Representative: 39.76% attrition rate.
- Laboratory Technician: 23.94% attrition rate.
- Human Resources job role: 23.08% attrition rate.
- Sales Executive: 17.48% attrition rate.
- Research Scientist: 16.10% attrition rate.
- Overtime = No: 10.44% attrition rate.
- Overtime = Yes: 30.53% attrition rate.
- Salary band Below 3K: 28.61% attrition rate.
- Salary band 3K–6K: 12.72% attrition rate.
- Salary band 6K–10K: 12.00% attrition rate.
- Salary band 10K+: 8.90% attrition rate.

These patterns indicate that overtime, lower income bands, certain job roles, and specific departments are associated with comparatively higher observed attrition in the dataset. These are analytical associations and should not automatically be interpreted as causal relationships.

15. Data Dictionary – Key Fields

Field	Business Description	Data Type
EmployeeNumber	Unique employee identifier	Integer
Age	Employee age	Integer
Attrition	Employee exit status; 0 = Active, 1 = Left	Binary 0/1
Department	Organizational division	Text

JobRole	Specific job role/title	Text
MonthlyIncome	Employee monthly income	Integer
OverTime	Whether employee works overtime	Text
JobSatisfaction	Job satisfaction rating	Ordinal 1–4
WorkLifeBalance	Work-life balance rating	Ordinal 1–4
BusinessTravel	Business travel frequency	Text
YearsAtCompany	Tenure at organization	Integer
EducationField	Employee education field	Text
JobLevel	Employee job level	Integer
JobInvolvement	Employee involvement level	Ordinal
PerformanceRating	Employee performance rating	Ordinal

16. Tools & Technologies

- Microsoft SQL Server
- SQL / CTEs / Window Functions
- Python / Pandas / NumPy / Matplotlib / Seaborn
- Jupyter Notebook
- Power BI / DAX
- Tableau
- Git / GitHub

17. Skills Demonstrated

- Data Cleaning and Validation
- Exploratory Data Analysis
- Advanced SQL
- CTEs and Window Functions
- Data Aggregation
- KPI Development
- DAX
- Tableau Calculated Fields
- Data Visualization
- Dashboard Development
- Interactive Filtering
- Dashboard Navigation
- Business Analysis
- HR Analytics
- Data Storytelling
- Cross-Platform BI Development

18. Project Structure

```
HR_Analytics_Portfolio_Project/
|
├── 1_Data/
|   ├── raw_data.csv
|   └── cleaned_data.csv
|
├── 2_SQL/
|   └── HR_Analytics_SQL.sql
|
├── 3_Python/
|   └── HR_EDA.ipynb
|
├── 4_PowerBI/
|   └── HR_Analytics_Dashboard.pbix
|
├── 5_Tableau/
|   └── HR_Analytics_Dashboard.twbx
|
├── 6_Documentation/
|   └── HR_Analytics_Project_Documentation.docx
|
├── 7_Images/
|   ├── PowerBI_Dashboard_1.png
|   ├── PowerBI_Dashboard_2.png
|   ├── PowerBI_Dashboard_3.png
|   ├── Tableau_Dashboard_1.png
|   └── Tableau_Dashboard_2.png
|
└── README.md
```

19. Source Consistency Note

The cleaned CSV and Tableau workbook use Attrition as a numeric 0/1 field. The Power BI Data Dictionary page contains a DAX example referencing "Yes". This represents a documentation mismatch in the supplied Power BI export.

For the final analytical dataset: 0 = Active; 1 = Left. The SQL, Python, and Tableau analysis uses the verified numeric representation.

20. Project Outcome

The completed solution demonstrates a full analytics pipeline from raw data preparation through SQL, Python EDA, Power BI reporting, and Tableau dashboard development.

- Workforce composition
- Employee attrition
- Compensation
- Job satisfaction
- Work-life balance
- Business travel
- Employee tenure
- Department-level attrition
- Job-role-level attrition
- Overtime-related attrition patterns

The project demonstrates the ability to apply consistent analytical logic across SQL, Python, Power BI, and Tableau, making it suitable as a portfolio project for Data Analyst and Business Analyst roles.

21. Final Project Files

- HR_Analytics_SQL.sql
- HR_EDA.ipynb
- raw_data.csv
- cleaned_data.csv
- HR_Analytics_Dashboard.pbix
- HR_Analytics_Dashboard.twbx
- Power BI dashboard screenshots
- Tableau dashboard screenshots
- Project Documentation
- README.md

Author

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Data Analytics | SQL | Python | Power BI | Tableau | Excel